europass Curriculum vitae

PERSONAL INFORMATION

Md Sakib Hasan Emon

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github.com/EMON-26

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Nationality Bangladeshi

PERSONAL STATEMENT

Computer Science and Engineering graduate with a strong academic focus on Machine Learning, Data Mining, Deep Learning, and Natural Language Processing. Experienced in data collection, preprocessing, feature engineering, model training, and evaluation using Python-based ML frameworks. Actively engaged in research on mental health prediction from Bangla–English code-mixed (Banglish) social media data. Seeking a fully funded Master's programme in Artificial Intelligence, Machine Learning, or Data Science to advance research in Natural Language Processing and Explainable AI.

WORK EXPERIENCE

2025 – 2025 **Software QA Intern**
QA Harbour

– Executing structured manual and functional test cases across web-based applications.
– Documenting test scenarios, identifying and reporting defects, and supporting regression testing cycles.

Business or sector Information Technology – Software Quality Assurance

2025 **Software Testing & QA Contributor**
UNITY AID – Integrated Design Project (IDP), Green University of Bangladesh

– Designed and executed manual test cases; performed functional and usability testing across all major user flows.
– Reported defects systematically, verified role-based access control and form validation, improving platform reliability.

Business or sector Higher Education – Integrated Design Project

EDUCATION AND TRAINING

2022 – 26 April 2026 **B.Sc. in Computer Science & Engineering**
Green University of Bangladesh
Grade CGPA 3.47 / 4.00

2018 – 2020 **Higher Secondary Certificate (HSC)** – Science
Mymensingh Govt. College
Grade GPA 5.00 / 5.00

2018 **Secondary School Certificate (SSC)** – Science
Chattogram Cantonment Board High School
Grade GPA 5.00 / 5.00

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RESEARCH EXPERIENCE

2025 – 26 April 2026

Bachelor's Thesis

Banglish Mental Health Classification using Transformer-Based Multi-Task Learning

Department of Computer Science and Engineering, Green University of Bangladesh

- Conducted undergraduate research on automatic mental health classification from Bangla-English (Banglish) code-mixed social media text.
- Constructed and annotated a large-scale Banglish mental health dataset containing more than 56,000 samples covering six mental health categories and three severity levels.
- Developed a Transformer-based multi-task learning framework using BanglaBERT with Attention Pooling for simultaneous mental health classification and severity prediction.
- Applied Explainable Artificial Intelligence (Grad×Input) to improve model interpretability and support transparent clinical decision making.
- Achieved 91.4% classification accuracy (Macro-F1: 0.918) for six-class mental health prediction and 92.2% accuracy (Macro-F1: 0.926) for severity estimation.
- Research Interests: Machine Learning, Natural Language Processing, Explainable AI, Healthcare AI, Low-Resource NLP, Large Language Models.

PUBLICATIONS

2026

Mental Health Classification from Social Media Data in Banglish using Transformer-Based Multi-Task Learning and Explainable AI

Authors: Md Sakib Hasan Emon, Md. Jubayer, Maruf Hasan Shuvo, Md. Solaiman Mia, and Md. Shoab Alam.

Conference: 11th IEEE Asia-Pacific Conference on Computer Science and Data Engineering (IEEE CSDE 2026), Cox's Bazar, Bangladesh, 1–3 November 2026.

Status: Accepted for presentation. Paper ID: ieee-csde_3742. Publication in the conference proceedings / IEEE Xplore pending.

PROJECTS

Banglish Mental Health Classification using Transformer-Based Multi-Task Learning

Python | NLP | Machine Learning | XAI | Research

GitHub: github.com/EMON-26

- Developed an end-to-end NLP framework for multi-class mental health classification and severity prediction from Bangla-English (Banglish) code-mixed social media text.
- Constructed and processed a dataset of 56,522 annotated Banglish samples covering six mental health categories and three severity levels.
- Implemented a BanglaBERT-based multi-task learning architecture with Attention Pooling and Grad×Input Explainable AI for interpretable predictions.
- Achieved **91.4%** classification accuracy (Macro-F1: **0.918**) and **92.2%** accuracy (Macro-F1: **0.926**) for severity prediction.

UNITY AID – Community Emergency Response Platform Software Engineering | Web Platform | QA

GitHub: github.com/EMON-26/unity-aid-testing

- Developed a web-based emergency response platform supporting volunteer coordination and verified case management.
- Implemented role-based authentication, emergency request management and location-aware service workflows.
- Designed and executed comprehensive functional, usability and regression testing to improve software reliability.
- Produced software testing documentation including test cases, bug reports and validation reports.

Employee Management System Java | Desktop Application

GitHub: github.com/EMON-26

- Developed a Java Swing desktop application for employee information management.
- Implemented CRUD operations with efficient record searching and reporting features.
- Designed a user-friendly graphical interface for administrative tasks.

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Inventory Management System Bash | Linux
GitHub: github.com/EMON-26

- Developed a menu-driven inventory management application using Bash scripting.
- Implemented inventory storage, searching, updating and reporting functionalities.
- Applied shell scripting techniques to automate routine inventory management tasks.

PERSONAL SKILLS

Mother tongue: Bangla

Other languages

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken interaction	Spoken production	
English	B2	B2	B2	B2	B2

Levels: A1 and A2: Basic user – B1 and B2: Independent user – C1 and C2: Proficient user
Common European Framework of Reference for Languages

Communication skills

- Strong technical writing and academic documentation skills.
- Experience presenting machine learning research findings and collaborating in multidisciplinary project teams.
- Effective written and verbal communication in English and Bangla.

Organisational / managerial skills

- Coordinated software testing and quality assurance activities for the UNITY AID project.
- Experience managing research workflows, dataset preparation and experimental evaluation.
- Former member of the Green University Debate Club.
- Former Cadet, Bangladesh National Cadet Corps (BNCC).

Technical skills

Programming	Python, Java, C
Machine Learning	Scikit-learn, PyTorch, TensorFlow, Transformers
Data Science	Pandas, NumPy, Matplotlib, NLTK, SHAP
Tools	Git, GitHub, Linux, Bash, LaTeX, Streamlit

Awards & Certifications

- BNCC Annual Training Encampment (ATE) Certificate
- BNCC Basic Training Encampment (BTE) Certificate
- Science Fair Participation Certificate
- Recognition for Contributions to the UNITY AID Integrated Design Project

Extracurricular activities

- Former Member – Green University Debate Club
- Former Cadet – Bangladesh National Cadet Corps (BNCC), Secondary School Level

Professional Profiles

Portfolio: `https://mdsakibhasanemon.netlify.app`
GitHub: `https://github.com/EMON-26`
LinkedIn: `https://linkedin.com/in/mdsakibhasan26`

References

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